

Decision Making Under Uncertainty

Homework 2

Kyle · ENGS 177 · Prof. Marrero · May 1, 2026

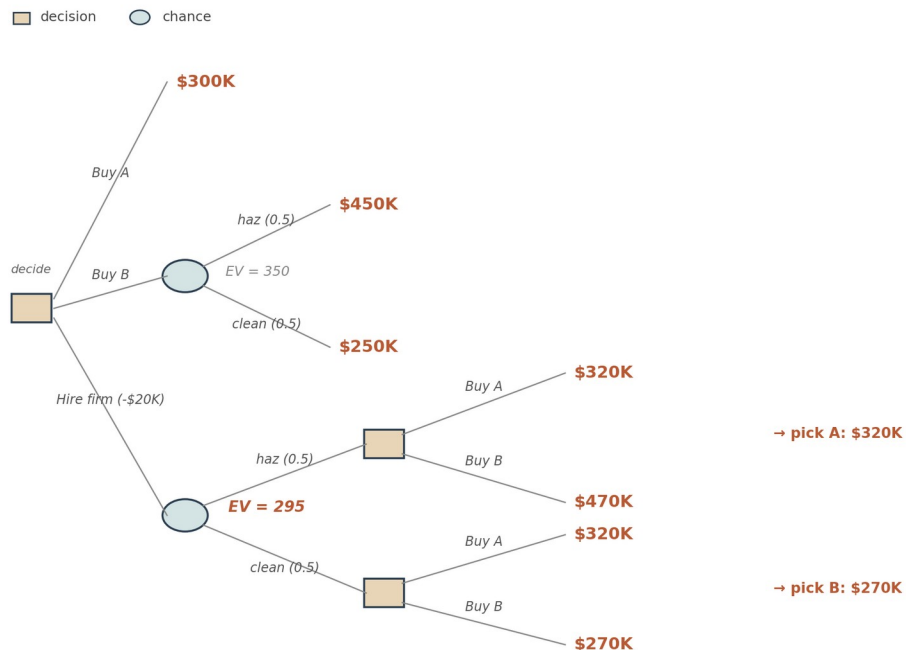
A note on AI use. Claude (Anthropic) helped me catch a handful of arithmetic slips, pressure-test the MDP formulations when I was unsure, and format this document. The solutions and the reasoning behind them are mine.

01 Hotel Lot Decision

Two lots are on the table. Lot A costs \$300K with no environmental concerns. Lot B costs \$250K but might be hazardous, and if it is, the cleanup is another \$200K. Prior on the hazard is 50/50. Before committing, the client can pay an environmental firm \$20K for a perfect test.

All outcomes are expressed as total cash paid by the client (so we minimize).

a. Decision tree



Squares are decision nodes; circles are chance nodes. Numbers on the right are total cash paid in \$K. The two italicized *EV* labels show the rolled-back expected value at each chance node, computed below.

b. Optimal policy via expected utility rollback

Since the client only cares about money paid (more is worse), I work directly with expected dollar cost rather than constructing a separate utility. Roll back from leaves to root.

Branch 1: Buy A directly.

$$\text{Cost} = 300.$$

Branch 2: Buy B directly.

$$E[\text{cost}] = 0.5 * 450 + 0.5 * 250 = 350.$$

Branch 3: Hire firm, then decide based on the test result. First, fold up the two sub-decisions:

Test result	Buy A cost	Buy B cost	Optimal action	Rolled-back cost
Hazardous	320	470	Buy A	320
Clean	320	270	Buy B	270

Then average over the test outcome:

$$E[\text{cost} \mid \text{hire firm}] = 0.5 * 320 + 0.5 * 270 = 295.$$

Compare the three top-level branches:

Action	E[total cost in \$K]
Buy A	300
Buy B	350
Hire firm, then act on result	295

Answer. Optimal policy: hire the testing firm. If the test says hazardous, buy A; if clean, buy B. Expected total cost is \$295K, beating Buy A (\$300K) and Buy B (\$350K).

c. Maximum willingness to pay for the testing firm

This is the expected value of perfect information. The cleanest way to compute it: imagine the test were free, and compare to the best you could do with no information.

Best decision with no information. Buy A, expected cost \$300K.

Best decision with free perfect information.

$$E[\text{cost} \mid \text{free test}] = 0.5 * (\text{cost if clean: Buy B} = 250) + 0.5 * (\text{cost if haz: Buy A} = 300) = 275.$$

$$\text{EVPI} = \$300\text{K} - \$275\text{K} = \$25\text{K}.$$

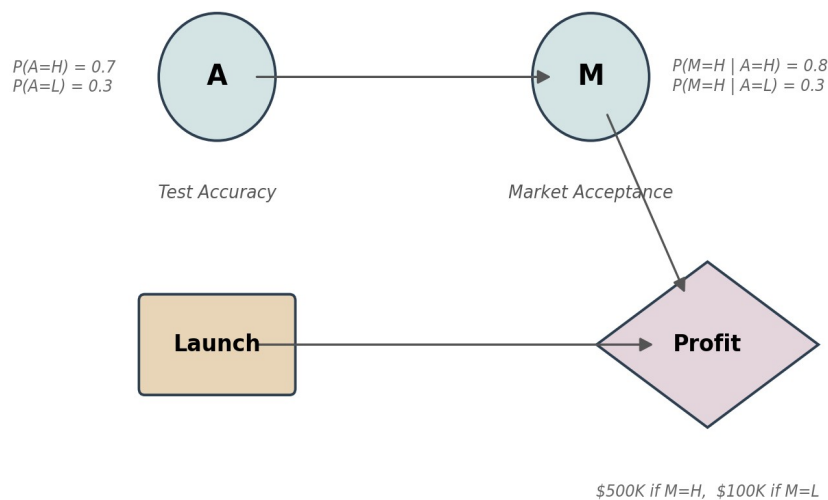
Answer. The most the client should pay for the test is \$25K. The actual price is \$20K, which is below that ceiling, so hiring the firm is worthwhile (consistent with part b).

Worth noting: the gap between actual cost (\$20K) and the ceiling (\$25K) is exactly the value the client captures by hiring the firm at this price, namely \$300K - \$295K = \$5K of expected savings.

02 Diagnostic Test Influence Diagram

a. Influence diagram

Three random/value nodes plus the launch decision. The accuracy A is exogenous and feeds the market acceptance M , which together with the launch decision determines profit.



Edges: $A \rightarrow M$ (accuracy influences acceptance), $M \rightarrow Profit$ (acceptance determines payoff), and $Launch \rightarrow Profit$ (no launch means no profit). The information arc into the decision node is empty: the company decides whether to launch before observing either A or M , so the decision is based solely on prior expectations.

b. Expected profit from launching

Marginalize accuracy out of acceptance first, then weight payoffs.

Step 1: marginal probability of high acceptance.

$$\begin{aligned} P(M=H) &= P(M=H | A=H) * P(A=H) + P(M=H | A=L) * P(A=L) \\ &= 0.8 * 0.7 + 0.3 * 0.3 = 0.56 + 0.09 = 0.65. \\ P(M=L) &= 1 - 0.65 = 0.35. \end{aligned}$$

Step 2: expected profit.

$$\begin{aligned} E[\text{Profit} \mid \text{Launch}] &= 0.65 * 500,000 + 0.35 * 100,000 \\ &= 325,000 + 35,000 = 360,000. \end{aligned}$$

Answer. Expected profit from launching is \$360,000. Since this is positive (and the implicit alternative of not launching yields \$0), the diagram recommends launching.

A structural note worth making. If accuracy were perfectly informative of acceptance ($P(M=H \mid A=H) = 1$, $P(M=H \mid A=L) = 0$), the marginal $P(M=H)$ would equal $P(A=H) = 0.7$ and expected profit would rise to $0.7(500K) + 0.3(100K) = \$380K$. The actual \$360K reflects the noise in the accuracy-to-acceptance link, which costs about \$20K of expected value.

03 Parking MDP

Each school day brings a parking choice and, possibly, a follow-up decision about whether to repair. The objective is to minimize the long-run expected average cost per day.

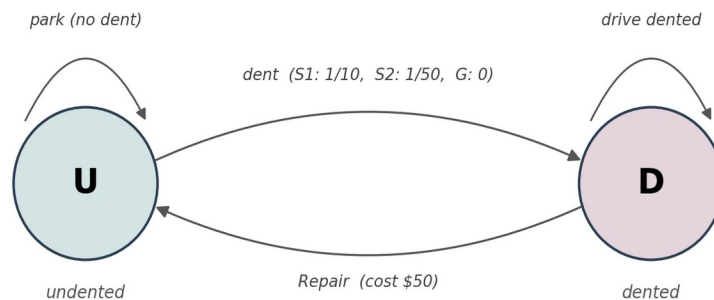
a. MDP formulation

State space. Two states capture everything that matters for the next decision:

- U : car is undented (or freshly repaired and back in service).
- D : car is dented and currently being driven.

Action space depends on the state.

- In U : $A(U) = \{S1, S2, G\}$, where $S1$ = street, one space; $S2$ = street, two spaces; G = garage.
- In D : $A(D) = \{R, dS1, dS2, dG\}$, where R = repair the car, and $dS1/dS2/dG$ mean drive dented and park in the corresponding spot.



Transition probabilities.

State, Action	$P(\rightarrow U)$	$P(\rightarrow D)$
U, S1	9/10	1/10
U, S2	49/50	1/50
U, G	1	0
D, R	1	0
D, dS1 / dS2 / dG	0	1

Cost function (expected one-step cost).

State, Action	Expected cost (\$)
U, S1	0
U, S2	$(3/10) * 15 = 4.50$
U, G	5
D, R	50
D, dS1	9
D, dS2	$9 + 4.50 = 13.50$
D, dG	$9 + 5 = 14$

Objective. Find a stationary policy π minimizing the long-run expected average cost per day:

$$g^* = \min_{\pi} \lim_{T \rightarrow \infty} (1/T) * E_{\pi} [\sum_{t=0}^{T-1} c(S_t, A_t)].$$

Bellman equation for average cost. Let h be the differential value function. Then for each state s :

$$g + h(s) = \min_a \{ c(s, a) + \sum_{s'} P(s' | s, a) * h(s') \}.$$

b. Solving by steady-state cycle analysis

Rather than solving Bellman directly, I can use the fact that any policy in this MDP is determined by two choices: where to park in U and what to do in D . A repair-and-restart cycle is the natural unit.

Claim: it never pays to drive dented forever. Driving dented locks in \$9/day plus parking, with no chance of returning to U , so the long-run cost is at least \$9/day. Repair returns the car to U and resets the cycle. So the optimal action in D is R (repair) for any policy that beats \$9/day, which all of them will.

Cycle structure under $(\pi(U), R)$. Each cycle has a geometric number of days in U before a dent transition, followed by exactly one day in D with the repair cost. If the dent probability is p and the cost in U is c , then expected days in U per cycle is $1/p$, expected cost in U per cycle is c/p , and

$$\text{avg cost per day} = (c/p + 50) / (1/p + 1) = (c + 50p) / (1 + p).$$

Plug in each parking option:

Park in U	c	p	avg cost (\$/day)
S1	0	1/10	$(0 + 5) / (1 + 0.1) = 50/11 \approx 4.55$

S2	4.50	1/50	$(4.5 + 1) / (1 + 0.02) = 275/51 \approx 5.39$
G	5	0	5 (no dents, no repairs)

Answer. Optimal policy: park in one street space (S1) when undented; repair when dented. Long-run average cost is $50/11 \approx \$4.55$ per day.

The intuition is appealingly counterintuitive. Taking two spaces costs you a near-guaranteed ticket and barely lowers the dent risk; the garage is dent-proof but expensive. The cheap S1 option, accepting an occasional \$50 repair, beats both.

04 Tic-Tac-Toe (3 pieces each) as an MDP

Standard tic-tac-toe gives each player as many pieces as they need; here the rules are tightened: each player has exactly 3 pieces. After all 3 are placed, on subsequent turns the player picks up one of her own pieces and re-places it on a free cell. First to get 3-in-a-row wins. Modeling from one player's view (call her X), the opponent's policy is folded into the environment.

a. Decision epochs and time horizon

Decision epochs are X's turns. Between consecutive X turns, O moves once, which is treated as part of the stochastic transition. The horizon is episodic: the game ends when someone wins, or when a step-limit T is reached (with the latter scored as a draw to guarantee termination, since piece-shuffling games can in principle cycle). I take $T = 50$ half-moves as a generous upper bound.

b. State space

A state is a tuple $s = (\text{board}, \text{whose_turn}, \text{status})$, where the board is a function from the 9 cells to $\{\text{empty}, X, O\}$ subject to two constraints: at most 3 X pieces and at most 3 O pieces. The *status* flag is one of $\{\text{ongoing}, X_wins, O_wins, \text{draw}\}$ and identifies terminal states.

Size of the state space. Counting non-terminal X-to-move boards by the number of pieces of each color (n_x, n_o):

$$|S| = \sum_{n_x=0..3} \sum_{n_o=0..3} C(9, n_x) * C(9 - n_x, n_o).$$

This sums to a few thousand legal board configurations, easily small enough for tabular methods.

c. Action space

Action set depends on how many X pieces are already on the board:

- If $n_x < 3$ (placement phase): $A(s) = \{ \text{place } X \text{ at cell } c : c \text{ is empty} \}.$
- If $n_x = 3$ (movement phase): $A(s) = \{ (c_1, c_2) : c_1 \text{ holds an } X \text{ piece, } c_2 \text{ is empty} \}.$

In the placement phase, $|A(s)| \leq 9$; in the movement phase, $|A(s)| \leq 3 * 6 = 18$.

d. Transition probabilities

Let $apply(s, a)$ denote the deterministic board after X plays a . Two cases:

- **X's move ends the game.** If $apply(s, a)$ places three X pieces in a line, transition deterministically to the X_wins terminal.
- **O moves next, stochastically.** Otherwise the next state is determined by O's response under some opponent model $\pi_O(\cdot | s')$:

$$P(s'' | s, a) = \sum_{a_O} \pi_O(a_O | apply(s, a)) * I(s'' = apply(apply(s, a), a_O)).$$

If $apply(apply(s, a), a_O)$ is a 3-in-a-row for O, the next state is the O_wins terminal. The opponent model π_O is a modeling choice. A common starting point is uniform-random over legal moves; a more realistic baseline would be a one-ply minimax opponent. In either case the opponent's policy is treated as fixed during planning, so the problem is genuinely an MDP rather than a game-theoretic one.

e. Reward function

Sparse terminal reward, zero everywhere else:

$$R(s, a, s'') = +1 \text{ if } s'' = X_wins; -1 \text{ if } s'' = O_wins; 0 \text{ otherwise.}$$

Draws receive reward 0. To encourage faster wins, an alternative is -0.01 per non-terminal step, but I'll keep the simpler sparse form for the formulation.

f. Discount factor and objective

Use $\gamma = 1$ since the horizon is bounded by T , or $\gamma \in (0, 1)$ if shorter wins should be preferred. The objective is to find a policy $\pi_O: S \rightarrow A$ maximizing

$$V^*(s_0) = \max_{\pi} E_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t * R(s_t, a_t, s_{t+1}) \mid s_0 \right].$$

Standard solution methods apply: tabular value iteration is feasible given the modest state space, and Q-learning against the chosen opponent model gives a model-free alternative if π_O is implicit (e.g., a fixed neural opponent).

05 Term Project Formulation

Regime-switching asset allocation. This builds on the preliminary description from Homework 1. Here I commit to specific elements.

a. Modeling choice

The model is a finite-horizon Partially Observable Markov Decision Process, with the partial observability handled by an HMM filter that supplies the belief state. Briefly, why this is the right object and not the alternatives the prompt mentions:

- **Decision tree.** A decision tree is good for short-horizon problems with a small set of branches, where outcomes resolve quickly and the tree can be drawn out. My horizon is $T = 60$ monthly decisions; a fully-enumerated tree with 4 actions and stochastic returns at each node is intractable by inspection.
- **Influence diagram.** Influence diagrams handle one-shot decisions or a small fixed sequence of decisions cleanly, but they don't natively encode a long Markov-style decision sequence with belief evolution. I would have to unroll into a separate node per period, which is just an MDP in disguise.
- **Plain MDP.** An MDP assumes the state is observable. The market regime in this problem is latent, so a plain MDP does not apply. The POMDP extension handles exactly this case by replacing the unobservable state with a belief over states.

b. Model elements

Decision epochs and horizon. Monthly rebalances over a 5-year window: $t = 0, 1, \dots, T-1$ with $T = 60$.

Hidden state. $s_t \in S = \{Expansion, Stress\}$. A 3-state extension $\{Expansion, Transition, Stress\}$ will appear as a robustness check.

Observations. At each period the manager observes $o_t = (VIX_t, TermSpread_t, CreditSpread_t) \in \mathbb{R}^3$, with regime-conditional emission $o_t | s_t \sim N(\mu_{\{s_t\}}, \Sigma_{\{s_t\}})$.

Belief state. $b_t \in \Delta(S)$, a probability vector over regimes. Since $|S| = 2$, b_t is parameterized by a single scalar $p_t = P(s_t = Stress | o_{1:t})$, updated by the forward algorithm:

$$b_{t+1}(s') \propto P(o_{t+1} | s') * \sum_s P(s' | s) * b_t(s).$$

Action space. $a_t \in A = \{0.20, 0.40, 0.60, 0.80\}$, the equity weight (bond weight is $1 - a_t$).

Transition. A stationary regime transition matrix $T(s' | s)$ estimated by Baum-Welch on the macro-financial observation series:

$$T = [[P(E \rightarrow E), P(E \rightarrow S)], [P(S \rightarrow E), P(S \rightarrow S)]].$$

Empirically, regime transitions are sticky: monthly $P(\text{stay in current regime})$ is typically 0.95–0.99 in macro-financial HMMs, giving expected regime durations on the order of 1–3 years.

Asset returns. Conditional on the regime, monthly equity and bond returns are drawn from regime-specific normal distributions, with non-zero correlation that itself shifts across regimes (for example, the equity-bond correlation flips sign between Expansion and Stress regimes in much of the post-2000 sample):

$$r_{eq,t} | S_t \sim N(\mu_{eq,s}, \sigma_{eq,s}), \quad r_{bd,t} | S_t \sim N(\mu_{bd,s}, \sigma_{bd,s}).$$

Portfolio return:

$$r_p(a_t, S_t) = a_t * r_{eq,t} + (1 - a_t) * r_{bd,t}.$$

c. Objective and utility

The agent maximizes expected discounted utility of monthly portfolio returns. I use a constant-relative-risk-aversion (CRRA) utility, which has two properties I want: it is concave (capturing risk aversion) and it scales well with wealth (no anchoring on a particular dollar level).

$$u(r) = ((1 + r)^{1-\gamma} - 1) / (1 - \gamma), \quad \gamma \in \{2, 4, 6\} \text{ for sensitivity, baseline } \gamma = 4.$$

CRRA collapses to logarithmic utility $u(r) = \log(1 + r)$ in the limit $\gamma \rightarrow 1$, which I will run as one of the sensitivity cases.

Discount factor. $\beta = 0.997$ per month, corresponding to roughly 4% annual time discount.

Value function and Bellman equation. Defined over belief states:

$$V_t(b) = \max_a E_{s, r_{eq}, r_{bd}, o' | b} [u(r_p(a, s)) + \beta * V_{t+1}(\tau(b, a, o'))],$$

where $\tau(b, a, o')$ is the belief-update operator (the forward filter applied to the new observation). Boundary condition: $V_T(b) = 0$ for all b .

Solution method. Exact POMDP solution is intractable for continuous observations. The plan is to use one-step lookahead with the QMDP heuristic:

$$\pi(b) \approx \operatorname{argmax}_a \sum_s b(s) * Q_{MDP}(s, a),$$

where $Q_MDP(s, a)$ is computed by solving the underlying fully-observed MDP (treating each regime as known) via standard value iteration. QMDP is known to underweight the value of information-gathering, but for an allocation problem driven mostly by exploitation of belief rather than active probing, it tends to perform within a few percent of optimal in published tests, while running in seconds rather than days.

What I expect to find. My working hypothesis, before any backtesting, is that the optimal policy will tilt aggressively toward equities ($a_t = 0.80$) when the belief in Expansion is high, and step down through $0.60 \rightarrow 0.40 \rightarrow 0.20$ as belief in Stress rises. The interesting question is whether the resulting portfolio beats a static 60/40 benchmark on risk-adjusted return, or whether the model's gains on regime calls are eaten by transaction costs and parameter uncertainty in a real out-of-sample window.